SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716
Name : Gangisetty Nithin kumar
Reg No : 192325048

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
Sol:

Given

- Network Address = 192.168.20.0/24
- Lab A → /26
- Lab B → /27
- Lab C → /28

Method Used

Using Fixed Length Subnet Mask (FLSM/VLSM allocation from the given block).

Formula:

- Number of Hosts = $2^{(32-prefix)}$
- Usable Hosts = $2^{(32-prefi\ )} - 2$

Solution

Lab A (/26)

Subnet Mask

255.255.255.192

Block Size

64

Network Address

192.168.20.0

Host Range

192.168.20.1 – 192.168.20.62

Broadcast Address

192.168.20.63

Usable Hosts

$$2^6 - 2 = 64 - 2 = 62$$

Usable Hosts = 62

Lab B (/27)

Next available subnet begins from 192.168.20.64

Subnet Mask

255.255.255.224

Network Address

192.168.20.64

Host Range

192.168.20.65 – 192.168.20.94

Broadcast Address

192.168.20.95

Usable Hosts

$$2^5 - 2 = 32 - 2 = 30$$

Usable Hosts = 30

Lab C (/28)

Next available subnet begins from 192.168.20.96

Subnet Mask

255.255.255.240

Network Address

192.168.20.96

Host Range

192.168.20.97 – 192.168.20.110

Broadcast Address

192.168.20.111

Usable Hosts

$$2^4 - 2 = 16 - 2 = 14$$

Usable Hosts = 14

Final Result

Lab	Prefix	Network Address	Host Range	Broadcast Address	Usable Hosts
Lab A	/26	192.168.20.0	192.168.20.1 – 192.168.20.62	192.168.20.63	62
Lab B	/27	192.168.20.64	192.168.20.65 – 192.168.20.94	192.168.20.95	30
Lab C	/28	192.168.20.96	192.168.20.97 – 192.168.20.110	192.168.20.111	14

Interpretation

The /24 network has been successfully divided into three subnetworks. Each laboratory receives its own subnet with sufficient host addresses, reducing broadcast traffic and improving network management.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable hosts addresses available in each subnet.

Given

System 1

IP Address = 192.168.1.130

Subnet Mask = 255.255.255.192 (/26)

System 2

IP Address = 192.168.1.190

Subnet Mask = 255.255.255.192 (/26)

Method Used

Subnet block size

$$256 - 192 = 64$$

Subnets

0–63

64–127

128–191

192–255

Solution

System 1

IP

192.168.1.130

Falls inside

128–191

Network Address

192.168.1.128

Broadcast Address

192.168.1.191

Host Range

192.168.1.129 – 192.168.1.190

Usable Hosts

$$64 - 2 = 62$$

Hence,

192.168.1.130 belongs to this subnet.

System 2

IP

192.168.1.190

Falls inside

128–191

Network Address

192.168.1.128

Broadcast Address

192.168.1.191

Host Range

192.168.1.129 – 192.168.1.190

Usable Hosts

62

Hence,

192.168.1.190 also belongs to the same subnet.

Final Result

System	Network Address	Host Range	Broadcast Address	Valid IP?	Usable Hosts
System 1	192.168.1.128	192.168.1.129 – 192.168.1.190	192.168.1.191	Yes	62
System 2	192.168.1.128	192.168.1.129 – 192.168.1.190	192.168.1.191	Yes	62

Interpretation

Both systems belong to the same subnet because they share the same network address (192.168.1.128/26). Therefore, they can communicate directly without requiring a router.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

Given

Network

172.16.0.0/16

Departments

HR → /25

Sales → /26

IT → /27

Admin → /28

Method Used

Allocate subnets sequentially using the given subnet masks.

Solution

HR (/25)

Network

172.16.0.0

Host Range

172.16.0.1 – 172.16.0.126

Broadcast

172.16.0.127

Usable Hosts

126

Sales (/26)

Network

172.16.0.128

Host Range

172.16.0.129 – 172.16.0.190

Broadcast

172.16.0.191

Usable Hosts

62

IT (/27)

Network

172.16.0.192

Host Range

172.16.0.193 – 172.16.0.222

Broadcast

172.16.0.223

Usable Hosts

30

Admin (/28)

Network

172.16.0.224

Host Range

172.16.0.225 – 172.16.0.238

Broadcast

172.16.0.239

Usable Hosts

14

Remaining Unused Range

After Admin subnet,

Remaining addresses begin from

172.16.0.240

Remaining range

172.16.0.240 – 172.16.255.255

Final Result

Department	Prefix	Network Address	Host Range	Broadcast	Usable Hosts
HR	/25	172.16.0.0	172.16.0.1 – 172.16.0.126	172.16.0.127	126
Sales	/26	172.16.0.128	172.16.0.129 – 172.16.0.190	172.16.0.191	62
IT	/27	172.16.0.192	172.16.0.193 – 172.16.0.222	172.16.0.223	30
Admin	/28	172.16.0.224	172.16.0.225 – 172.16.0.238	172.16.0.239	14

Interpretation

The organization efficiently allocates different subnet sizes according to departmental requirements while preserving a large unused address space for future expansion.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

Given

Network

172.16.10.0/24

Subnet Mask

/27

IP Address

172.16.10.78

Second IP

172.16.10.95

Method Used

Block Size

$$256 - 224 = 32$$

Subnets

0

32

64

96

128

160

192

224

Solution

Total Number of Subnets

Borrowed bits

27−24=3

$$2^3 = 8$$

Total Subnets

8

Usable Hosts

$$2^5 - 2 = 30$$

Usable Hosts

30

Subnet of 172.16.10.78

78 lies between

64–95

Network Address

172.16.10.64

Host Range

172.16.10.65 – 172.16.10.94

Broadcast

172.16.10.95

heck IP 172.16.10.95

172.16.10.95 is

Broadcast Address

Therefore,It **does not represent a valid host**.

Final Result

Item	Value
Total Subnets	8
Usable Hosts/Subnet	30
Network Address	172.16.10.64
Host Range	172.16.10.65 – 172.16.10.94
Broadcast Address	172.16.10.95
Is 172.16.10.95 in same subnet?	Yes, but it is the broadcast address (not a usable host).

Interpretation

The IP address 172.16.10.78 belongs to subnet 172.16.10.64/27. The address 172.16.10.95 is the broadcast address of the same subnet and therefore cannot be assigned to any host.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Given IPv6 Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Prefix

/64

Method Used

Apply IPv6 compression rules:

- Remove leading zeros.
- Replace one continuous sequence of all-zero blocks with "::".

Solution

Compressed Address

2001:db8::ff00:42:8329

Expanded Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Network Prefix (/64)

First 64 bits represent the network.

Network Prefix

2001:0db8:0000:0000::/64

Host Portion

Remaining

64 bits

Possible Host Addresses

$$2^{64} = 18{,}446{,}744{,}073{,}709{,}551{,}616$$

Possible Host Addresses

18,446,744,073,709,551,616

Final Result

Item	Value
Original Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed Address	2001:db8::ff00:42:8329
Expanded Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Network Prefix	2001:0db8:0000:0000::/64
Host Bits	64
Total Host Addresses	18,446,744,073,709,551,616

Interpretation

IPv6 provides an extremely large address space. With a /64 prefix, the first 64 bits identify the network, while the remaining 64 bits are available for interface addressing, supporting approximately **18.4 quintillion** unique host addresses within a single subnet.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation